

Tutorial 11

$$(26) \quad 3 \frac{dy}{dx} + \sqrt{y} = e^{0.1x}$$

$$\frac{dy}{dx} = \frac{e^{0.1x} - \sqrt{y}}{3}$$

$$f(x, y) = \frac{e^{0.1x} - \sqrt{y}}{3}$$

$$y_{i+1} = y_i + f(x_i, y_i)h$$

$$y_1 = y_0 + f(x_0, y_0)h$$

$$= 5 + f(0.3, 5) \cdot 0.3$$

$$= 5 + \left(\frac{e^{0.1 \times 0.3} - \sqrt{5}}{3} \right) 0.3$$

$$= 5 + (1.03 - 2.24) \cdot 0.1$$

$$= 4.879 //$$

$$x_1 = x_0 + h = 0.3 + 0.3 = 0.6$$

$$y(0.6) = y_1 = 4.879 //$$

$$y_2 = y_1 + f(x_1, y_1)h$$

$$= 4.879 + f(0.6, 4.879) \cdot 0.3$$

$$= 4.879 + \left(\frac{e^{0.1 \times 0.6} - \sqrt{4.879}}{3} \right) 0.3$$

$$= 4.879 + (1.06 - 2.208) \cdot 0.1$$

$$= 4.764,$$

$$x_2 = x_1 + h = 0.6 + 0.3 = 0.9$$

$$y(0.9) = y_2 = 4.764 //$$

$$\frac{dy}{dx}(0.9) = \frac{e^{0.1x} - \sqrt{y}}{3}$$

$$= \frac{e^{0.1 \times 0.9} - \sqrt{4.764}}{3}$$

$$= \frac{2.109 - 2.18}{3}$$

$$= -0.363$$

→ (B)

$$(27) \quad v(t) = 200 \ln(1+t) - t$$

$$\frac{ds}{dt} = 200 \ln(1+t) - t$$

$$f(t, s) = 200 \ln(1+t) - t$$

$$s_{i+1} = s_i + f(t_i, s_i)h$$

$$s_1 = s_0 + f(t_0, s_0)h$$

$$= 0 + f(1, 0) \cdot 5$$

$$= 0 + (200 \ln(1+1) - 1) \cdot 5$$

$$= 1088.6 //$$

$$t_1 = t_0 + h = 1 + 5 = 7 //$$

$$s_2 = s_1 + f(t_1, s_1)h$$

$$= 1088.6 + f(7, 1088.6) \cdot 5$$

$$= 1088.6 + (200 \ln(1+7) - 7) \cdot 5$$

$$= 3133.1 //$$

$$t_2 = t_1 + h = 7 + 5 = 12 //$$

$$s(12) - s(2) = s_2 - s_1$$

$$= 3133.1 - 0 = 3133.1$$

→ (A)

$$(28) \quad 2 \frac{dy}{dx} + 3y = e^{-5x}$$

$$\frac{dy}{dx} = \frac{e^{-5x} - 3y}{2}$$

$$\frac{d^2y}{dx^2} = \frac{-5e^{-5x} - 3 \frac{dy}{dx}}{2}$$

$$\frac{d^2 y}{dx^2} = \frac{-5e^{5x} \cdot 3(e^{-5x} \cdot \frac{1}{2})}{2}$$

$$= \frac{-15e^{5x} \cdot 3e^{-5x} \cdot \frac{1}{2}}{4}$$

$$= \frac{-13e^{-5x} \cdot \frac{1}{2}}{4}$$

$$y_{i+1} = y_i + f(x_i, y_i)h + f'(x_i, y_i)h^2$$

$$y_{1.1} = y_1 + \frac{1}{2}(e^{-5x} - y_1) + \frac{1}{2}(-\frac{15e^{-5x}}{4} + \frac{1}{4}y_1)$$

→ (C) //

$$(29) \frac{d\theta}{dt} = -k(\theta - \theta_a)$$

$$\frac{d\theta}{dt} + k\theta = k\theta_a$$

$$\text{At } t \rightarrow \infty \rightarrow r = -k$$

$$\theta_h = Ae^{-kt}$$

$$(Dh)\theta = k\theta_a$$

$$\theta_p = \theta_a$$

$$\theta = \theta_h + \theta_p$$

$$\theta = \theta_a$$

$$\theta = \theta_h + \theta_p$$

$$= Ae^{-kt} + \theta_a$$

$$\theta_a = 72$$

$$\theta(0) = 85$$

$$\theta(1) = 78$$

$$\theta(2) = 72.6$$

D - time of death

$$85 = Ae^{-k \cdot 0} + 72 - (1)$$

$$78 = Ae^{-k \cdot 1} + 72 - (2)$$

$$72.6 = Ae^{-k \cdot 2} + 72 - (3)$$

When (1)

$$Ae^{-k \cdot 0} = 13 - (4)$$

When (2)

$$Ae^{-k} = 6 - (5)$$

When (3)

$$e^{2k} = 2.1667$$

$$k = \frac{1}{2} \ln(2.1667)$$

$$k = 0.25773$$

When (5)

$$A = 61.028^\circ F$$

When (3)

$$72.6 = 61.028e^{-0.25773 \cdot 2} + 72$$

$$61.028e^{-0.25773 \cdot 2} = 26.6$$

$$B = 3.2220 //$$

3.13 P.M.

→ (B) //

$$(30) \sin(x) = x - \frac{(x)^3}{3!} + \frac{(x)^5}{5!} - \dots$$

$$= x - \frac{8x^3}{6} + \frac{32x^5}{120} - \dots$$

$$= x - 1.333x^3 + 0.26667x^5 - \dots$$

Hence x^3 term is 0.26667

→ (D) //

$$(31) f(x+h) = f(x) + f'(x)h + \frac{f''(x)h^2}{2!} + \frac{f'''(x)h^3}{3!} + \dots$$

$$h = 7 - 3 = 4$$

$$f(3+4) = f(3) + f'(3)4 + \frac{f''(3)4^2}{2!} + \frac{f'''(3)4^3}{3!} + \dots$$

Since all derivatives higher the seconds are zeros.

$$f(7) = f(3) + f'(3)4 + \frac{f''(3)16}{2!}$$

$$= 6 + 8 \times 4 + \frac{11 \times 16}{2!}$$

$$= 126$$

→ (C) //

$$(32) y(x+h) = y(x) + y'(x)h + \frac{y''(x)h^2}{2!}$$

$$h = 0.2 - 0 = 0.2$$

$$y(0+0.2) = y(0) + y'(0) \times 0.2 + \frac{y''(0)0.2^2}{2!}$$

$$y(0) = 3$$

$$y'(x) = y^2 + 2$$

$$y'(0) = 3^2 + 2 = 11$$

$$y''(x) = 2y(y' + 1)$$

$$y''(0) = 2(3)^2(3+2)$$

$$= 78$$

$$y(0.2) = 3 + 11 \times 0.2 + \frac{78 \times 0.2^2}{2}$$

$$= 24.46$$

→ (C) //

$$(33) \cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\cos(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 4^n x^{2n}}{(2n)!}$$

→ (B) //

$$(34) \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

$$f(t) = \frac{2e^{-t^2}}{\sqrt{\pi}}$$

$$f(0) = \frac{2e^0}{\sqrt{\pi}} = \frac{2}{\sqrt{\pi}}$$

$$f'(t) = \frac{2e^{-t^2}}{\sqrt{\pi}} (-2t)$$

$$f'(0) = \frac{2e^0}{\sqrt{\pi}} \times 0 = 0$$

$$f''(t) = \frac{2e^{-t^2}}{\sqrt{\pi}} (-2t) + \frac{2e^{-t^2}}{\sqrt{\pi}} (-2)$$

$$f''(0) = \frac{2e^0(-2)}{\sqrt{\pi}} = \frac{-4}{\sqrt{\pi}}$$

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)h^2}{2!}$$

$$f(0+h) = f(0) + f'(0)h + \frac{f''(0)h^2}{2!}$$

$$f(h) = \frac{2}{\sqrt{\pi}} + 0 + \frac{-4}{\sqrt{\pi}} \frac{h^2}{2!}$$

$$= \frac{2}{\sqrt{\pi}} - \frac{2}{\sqrt{\pi}} h^2$$

$$\frac{2e^{-h^2}}{\sqrt{\pi}} = \frac{2}{\sqrt{\pi}} - \frac{2}{\sqrt{\pi}} h^2$$

$$erf(x) = \int_0^x \left(\frac{2}{\sqrt{\pi}} - \frac{2t^2}{\sqrt{\pi}} \right) dt$$

$$= \left[\frac{2}{\sqrt{\pi}} t - \frac{2}{\sqrt{\pi}} \frac{t^3}{3} \right]_0^x$$

$$= \frac{2x}{\sqrt{\pi}} - \frac{2x^3}{3\sqrt{\pi}}$$

$$erf(x) = \frac{2(2)}{\sqrt{\pi}} - \frac{2(2)^3}{3\sqrt{\pi}}$$

$$= -0.75225 //$$

→ (A) //

$$(35) R_n(x) = \frac{x^{n+1}}{(n+1)!} f^{(n+1)}(c), \quad n \geq 0, \quad 0 \leq c \leq x$$

$$R_n(0.1) = \frac{(0.1)^{n+1}}{(n+1)!} f^{(n+1)}(c)$$

$$|\sin(x)| \leq 1 \quad \text{and} \quad |\cos(x)| \leq 1$$

$$|f^{(n+1)}(c)| \leq 1$$

$$R_n(0.1) \leq \frac{(0.1)^{n+1} (1)}{(n+1)!}$$

$$= \frac{(0.1)^{n+1}}{(n+1)!}$$

So when if

$$R_n(0.1) \leq 10^{-6}$$

$$\frac{(0.1)^{n+1}}{(n+1)!} \leq 10^{-6}$$

$$(n+1)!$$

$$n \geq 4$$

→ (B) //